

Functional detrended fluctuation analysis (FDFA): working notes — the idea, the axioms, and the prediction–falsification record

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Abstract

These are the working notes behind the FDFA project. They record, in the order in which it happened, the question that started the project, the minimal theory built to make the question precise, the falsifiable predictions derived from that theory, and the outcome of each prediction — including one falsification that became the most useful structural insight of the program. The notes accompany the code repository that implements everything described here; the consolidated manuscript is a separate document. Nothing in these notes is required to use the code, but they explain *why* the code is organized the way it is.

1 The question

Classical DFA/MFDFA (Peng et al., 1994; Kantelhardt et al., 2002) computes, in each window v of scale s , the *variance* of the detrending residuals,

$$F^2(v, s) = \frac{1}{s} \sum_{i=1}^s (Y((v-1)s + i) - P_m^{(v)}(i))^2, \quad (1)$$

and aggregates the windows with q -th power means, $F_q(s) = (\frac{1}{2N_s} \sum_v [F^2(v, s)]^{q/2})^{1/q} \sim s^{h(q)}$. Essentially every ingredient of this pipeline has been varied in the literature — the detrending, the windowing, the correlation structure probed — except the fluctuation measure itself, which has remained the variance in every variant. The project starts from one question: *is the variance necessary, and what exactly would change if it were replaced?*

The answer developed below: for the scaling exponent the variance is irrelevant (any admissible dispersion functional gives the same exponent); what the choice controls is the statistical behavior of the estimator; and the decisive design choice turns out to sit one level up — in how windows are aggregated — which in turn yields a surrogate-free diagnostic of the origin (correlational versus distributional) of multifractality.

2 The construction

Definition 2.1 (Dispersion functional). $\Phi : \bigcup_{s \geq 1} \mathbb{R}^s \rightarrow [0, \infty)$, measurable on each $\mathbb{R}^s$, satisfying (A1) positive homogeneity, $\Phi(\lambda r) = \lambda \Phi(r)$ for $\lambda > 0$; (A2) symmetry, $\Phi(-r) = \Phi(r)$, and permutation invariance; (A3) nondegeneracy and continuity; (A4) sampling consistency: for every càdlàg $f : [0, 1] \rightarrow \mathbb{R}$, $\Phi(f(1/s), \dots, f(s/s))$ converges as $s \rightarrow \infty$ to a nondegenerate limit $\Phi_\infty(f)$ (càdlàg, not merely continuous, so that jump processes are covered).

Definition 2.2 (FDFA, two layers). $F_\Phi(v, s) = \Phi(r^{(v, s)})$ on the window residuals (inner layer), then $F_{\Phi, \mathcal{A}}(s) = \mathcal{A}(\{F_\Phi(v, s)\}_v)$ across windows (outer layer), and the exponent from the log-log slope. Members are written $\text{FDFA}_{\Phi, \mathcal{A}}$. Classical DFA is $\text{FDFA}_{\text{RMS}, \mathcal{A}_2}$; MFDFA is the slice $\{\text{FDFA}_{\text{RMS}, \mathcal{A}_q}\}_q$.

The organizing idea: **the inner functional decides what “fluctuation” means inside a window; the outer aggregator decides which windows dominate.** Implemented inner functionals (all normalized to Gaussian σ -consistency): RMS, L^p ($p = 1, 1/2$), MAD, quantiles $Q_{0.75}$, $Q_{0.90}$, IQR, Huber scale, trimmed standard deviation, L^∞ , entropy power $e^{\hat{h}}$. Analyzed and excluded: bounded functionals (kernel MMD, total variation, KL — no scaling law is possible), scale-invariant functionals, and reducible ones ($W_p(\cdot, \delta_0)$ is exactly L^p ; energy distance is L^1 -type). Entropies are admissible only after exponentiation.

3 What the theory says

Lemma 3.1 (Dilation equivariance). *For Π_s the $L^2[0, s]$ -projection onto polynomials of degree $\leq m$ and $(D_a f)(t) = f(at)$: $(I - \Pi_s)g = D_{1/s}(I - \Pi_1)D_s g$, and $(I - \Pi_s)$ commutes with positive scalars.*

Proposition 3.2 (Exponent universality). *Let X be H -sssi with càdlàg paths, Φ admissible, and let $Z_\Phi = \Phi_\infty((I - \Pi_1)X|_{[0,1]})$ have finite nonzero median. Then $s^{-H}F_\Phi(v, s) \rightarrow_d Z_\Phi$, hence $\text{med } F_\Phi(v, s) = C_\Phi s^H(1 + o(1))$, and the same for the mean under uniform integrability. The exponent is H for every admissible Φ .*

The proof is three lines given the lemma: residuals of the s -window are, in distribution, s^H times time-rescaled residuals of the unit window; homogeneity extracts s^H ; sampling consistency gives the limit. Consequence: replacing the variance never changes the estimated quantity, only the estimator’s statistics — the *license* for everything else.

Remark 3.3 (Finite-range correction of L^∞). The supremum is admissible, but the sampled maximum approaches the continuum supremum from below with a slowly varying deficit (in the iid caricature, a $\sqrt{2\log s}$ factor; dependence within the window slows but does not remove it), inflating finite-range slopes — more at small H . Confirmed numerically; leader-type normalization (Jaffard, 2004) is the cure.

Proposition 3.4 (Heavy tails). *For iid symmetric α -stable input, $0 < \alpha < 2$ (profile = $1/\alpha$ -self-similar stable walk): (i) $s^{-1/\alpha}F_\Phi(v, s) \rightarrow_d \Xi$ with regularly varying tail of index α ; (ii) for the q -mean outer layer with $q > \alpha$, $\mathbb{E}\Xi^q = \infty$ and the bifractal law $h(q) = 1/q$ of Nakao (2000) applies — in particular $h(2) = 1/2$ for every $\alpha < 2$: classical DFA is asymptotically blind to the self-similarity index of stable noise; (iii) a median outer layer recovers $1/\alpha$ with any admissible inner functional.*

Principle 3.5 (Trend absorption). An extreme increment enters the profile as a step; after detrending it is a perturbation of order M spread over the whole window — a shape distortion, not a point outlier. No translation-insensitive inner functional, robust or not, can reject it. Robustness against heavy tails is effective only at the outer, cross-window layer.

This principle was *not* part of the original plan; it is the post-mortem of a falsified prediction (P4 below), and it dictates both the design rule (robustify the outer layer) and the diagnostic that follows.

4 The discriminator

A q -mean with large $|q|$ is dominated by extreme windows, and a window can be extreme for two indistinguishable reasons: genuinely different local scaling (the multifractality one wants) or a heavy-tailed event (distributional luck). Quantiles of the window ensemble separate the two.

Definition 4.1 (Quantile-exponent profile). $\theta(\tau)$ is the log-log slope of the empirical τ -quantile of $\{F_\Phi(v, s)\}_v$ against s , over scales with $2N_s \gtrsim 50$.

If windows are exchangeable with a common scale factor (heavy-tailed monofractal), every quantile scales identically: θ flat, even when $\Delta h = h(-4) - h(4)$ is large. If window heterogeneity is scale-structured (genuine multifractality), θ varies with τ — decreasing for cascades. Verdict rule, with thresholds fixed on synthetic data before touching real data: $\Delta h > 0.15$ and $|\Delta\theta| \leq 0.08 \Rightarrow$ distributional; $\Delta h > 0.15$ and $|\Delta\theta| > 0.08 \Rightarrow$ genuine. This answers, from a single run and without surrogate ensembles, the question customarily answered by shuffling (Kantelhardt et al., 2002; Sadegh Movahed et al., 2006); it is the fixed- N , per-record counterpart of the asymptotic dichotomy of Kwapień et al. (2023).

5 The prediction ledger

Every experiment tested a prediction formulated in advance. The record, in full (details and figures in the repository):

	prediction	benchmark	outcome
P1	all admissible Φ give slope H	fGn, $H \in \{0.3, 0.5, 0.7\}$	confirmed (spread < 0.011)
P2	classical blind, robust recovers $1/\alpha$	stable, $\alpha \in \{1.2, 1.5, 1.8\}$	confirmed, strong form
P3	genuine width preserved by robust Φ	binomial cascade	confirmed (0.23–0.25 vs 0.215)
P4	robust inner Φ suppresses spurious width	stable $h(q)$	falsified $\rightarrow$ Principle 3.5
P5	$(\Delta h, \Delta \theta)$ separates the three cases	fGn / cascade / stable	confirmed
P6	discriminator correct with Gaussian tails	MRW	confirmed ($\Delta \theta = -0.12$)
P7	universality beyond fBm construction	ARFIMA(0, d , 0)	confirmed
P8	θ -verdict = shuffle verdict	five signal classes	confirmed (5/5 at $n = 2^{14}$)
P9	RMSE $\propto n^{-1/2}$; order- m bias floor	fGn, stable grid	confirmed (50 real./point)

Tab. 1: The prediction–falsification record of the project. P4 is reported with the same prominence as the confirmations: its failure mechanism (trend absorption) reorganized the framework and produced the discriminator.

Two caveats found along the way and kept: the discriminator needs $n \gtrsim 10^4$ under heavy tails (a stable record was misclassified at $n = 2^{12}$), and zero-inflated series (e.g. sunspot numbers at solar minima) blow up negative- q moments unless the lower scale cut exceeds the zero-run scale — with the default cut one obtains a spurious $\Delta h \approx 10$.

6 Reality check (OMNI2)

Applied to four heliophysical series with FDFA_{Huber,med}⁽²⁾ for exponents and the discriminator for origin: sunspot R reproduces the surrogate-based conclusion of [Sadegh Movahed et al. \(2006\)](#) (width $0.62 \rightarrow 0.01$ under shuffling; $\Delta \theta = -0.16$) from a single run; D_{st} and IMF B come out genuinely multifractal, with visible classical/robust exponent gaps flagging tail influence; and the IMF increments δB are a *mixed* case — shuffling retains $\Delta h_{\text{shuf}} = 0.15$ of 0.38, so roughly 40% of the width is distributional, refining the picture of [Macek and Wawrzaszek \(2011\)](#).

7 Open problems

(1) Median outer layer sacrifices the Legendre formalism: robust exponents or spectra, not both from one pipeline. (2) $\Delta \theta$ needs a calibrated significance scale and a formal three-way genuine/mixed/distributional verdict. (3) Rigor gaps: discretization in Proposition 3.2; tail constants in Proposition 3.4(i); median concentration under long-range dependent window ensembles; quantitative trend absorption; multiplicative-invariance version for cascades. (4) Dictionary to the p -leader formalism ([Leonarduzzi et al., 2016](#)).

Code

Everything above is implemented in the accompanying repository: `fdfa.py` (the two-layer estimator with pluggable functionals), `generators.py` (benchmarks with known ground truth), `run_validation.py` (the nine-experiment ledger of Table 1), `run_application.py` (the OMNI2 analysis). All figures and tables regenerate from two seeded commands in under one minute.

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